

1 Solution — GIAM 3.4

Problem 9

1.1 Problem

True or false: whenever an integer n is a divisor of the square of an integer, m^2 , it follows that n is a divisor of m as well. In symbols,

$$\forall n \in \mathbb{Z}, \forall m \in \mathbb{Z}, \quad n \mid m^2 \implies n \mid m.$$

Prove your answer.

1.2 Answer

False. $n = 4, m = 2$: $4 \mid 4 = m^2$ but $4 \nmid 2$.

1.3 Disproof

Disproof. Take $n = 4$ and $m = 2$, both integers. Then

$$m^2 = 4 = 4 \cdot 1, \quad \text{so} \quad n \mid m^2.$$

But $n \nmid m$: if $4 \mid 2$ there would be an integer k with $2 = 4k$, forcing $k = 1/2$, which is not an integer.

So the hypothesis holds and the conclusion fails, and the universal claim is false.



This is the **least** modulus that fails: exhaustive search confirms that $n = 2$ and $n = 3$ both satisfy the implication for every m .

1.4 The Picture

The exponent view explains not only that the claim is false but exactly which moduli it fails for, and supplies a witness for each of them.

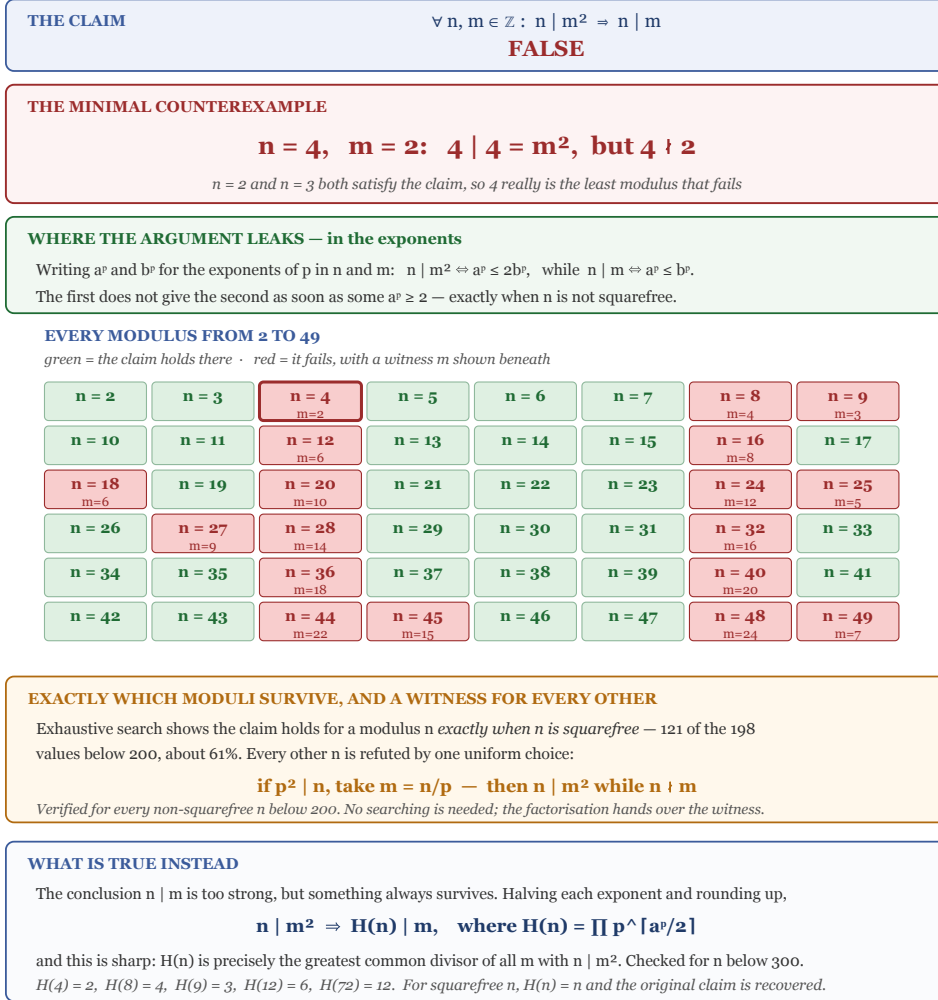


Figure 1.1: A diagram in five bands. The first states the claim — for all integers n and m , n divides m squared implies n divides m — and the verdict, FALSE. The second gives the minimal counterexample: n equals 4 and m equals 2, since 4 divides 4 which is m squared, but 4 does not divide 2. Both n equals 2 and n equals 3 satisfy the claim, so 4 really is the least modulus that fails. The third band shows where the argument leaks, in the exponents: writing a^p and b^p for the exponents of the prime p in n and in m , n divides m squared exactly when a^p is at most twice b^p , while n divides m exactly when a^p is at most b^p ; the first does not give the second as soon as some a^p is at least 2, which is exactly when n is not squarefree. The centre is a strip of every modulus from 2 to 49, one cell each. Green cells, where the claim holds, are the squarefree moduli. Red cells, where

it fails, are the non-squarefree ones, each labelled with a witness m : n equals 4 with m equals 2, n equals 8 with m equals 4, n equals 9 with m equals 3, n equals 12 with m equals 6, n equals 16 with m equals 8, n equals 18 with m equals 6, and so on. The cell for n equals 4 is outlined heavily. A band below states exactly which moduli survive and gives a witness for every other: exhaustive search shows the claim holds for a modulus n exactly when n is squarefree — 121 of the 198 values below 200, about 61 percent — and every other n is refuted by one uniform choice, namely if p squared divides n take m equal to n over p , whereupon n divides m squared while n does not divide m . This was verified for every non-squarefree n below 200; no searching is needed, since the factorisation hands over the witness. The final band gives what is true instead: the conclusion n divides m is too strong, but something always survives. Halving each exponent and rounding up, n divides m squared implies H of n divides m , where H of n is the product of p raised to the ceiling of a sub p over 2. This is sharp: H of n is precisely the greatest common divisor of all m with n dividing m squared, checked for n below 300. H of 4 is 2, H of 8 is 4, H of 9 is 3, H of 12 is 6, H of 72 is 12. For squarefree n , H of n equals n and the original claim is recovered.

1.5 Remark — Where the Argument Leaks

The cleanest way to see both the failure and its extent is through prime exponents. Write a_p and b_p for the exponent of p in n and in m . Then divisibility is a comparison of exponents:

$$n \mid m^2 \iff a_p \leq 2b_p \text{ for every } p, \quad n \mid m \iff a_p \leq b_p \text{ for every } p.$$

So the claim asserts

$$a_p \leq 2b_p \implies a_p \leq b_p,$$

which is false in general — it says a number bounded by *twice* something is bounded by the something. It survives only when a_p is too small for the doubling to matter: if $a_p \leq 1$ then $a_p \leq 2b_p$ forces $b_p \geq 1 \geq a_p$. As soon as some $a_p \geq 2$, choosing b_p strictly between $a_p/2$ and a_p breaks it.

The minimal instance is $a_p = 2$, $b_p = 1$ with $p = 2$ — that is, $n = 4$ and $m = 2$.

1.6 Remark — Exactly Which Moduli Survive

Exhaustive search over $2 \leq n < 200$ shows the implication holds for a modulus n **exactly when n is squarefree**, matching the exponent analysis precisely: squarefree means every $a_p \leq 1$.

That is **121** of the **198** moduli, about **61%** — consistent with the known density $6/\pi^2 \approx 0.6079$ of squarefree integers.

Moreover, every failing modulus has a witness that can be written down without any search:

If $p^2 \mid n$, take $m = n/p$. Then $m^2 = n^2/p^2$, and $n \mid n^2/p^2$ precisely because $p^2 \mid n$; while $n \nmid m$ since $0 < m < n$.

Verified for every non-squarefree $n < 200$. A few instances:

n	p	$m = n/p$	m^2
4	2	2	4
8	2	4	16
9	3	3	9
12	2	6	36
50	5	10	100

So the failure is not sporadic. It is structural, it affects roughly two moduli in five, and the factorisation of n hands over the counterexample directly.

1.7 Remark — What Is True Instead

A false conjecture is more instructive when one asks what the correct statement is. The conclusion $n \mid m$ is too strong, but the hypothesis is not empty. Halving each exponent and rounding up gives

$$n \mid m^2 \implies H(n) \mid m, \quad \text{where} \quad H(n) = \prod_p p^{\lceil a_p/2 \rceil}.$$

The proof is immediate from the exponent inequality: $a_p \leq 2b_p$ gives $b_p \geq a_p/2$, and b_p is an integer, so $b_p \geq \lceil a_p/2 \rceil$.

This is **sharp**. Computing the greatest common divisor of all m satisfying $n \mid m^2$ returns exactly $H(n)$, for every $n < 300$ — so no larger divisor can be guaranteed. Some values:

$$H(4) = 2, \quad H(8) = 4, \quad H(9) = 3, \quad H(12) = 6, \quad H(72) = 12, \quad H(200) = 20.$$

Two things fall out. First, when n is squarefree every $\lceil a_p/2 \rceil = a_p = 1$, so $H(n) = n$ and the original claim is recovered — the theorem of 3.3 Problem 2. Second, a weaker but memorable consequence: since $\lceil a_p/2 \rceil \geq 1$ whenever $p \mid n$,

$$n \mid m^2 \implies \text{rad}(n) \mid m,$$

where $\text{rad}(n)$ is the product of the distinct primes dividing n . Verified across all $n < 200$ and $m < 600$. In words: **m must contain every prime of n , just not necessarily to the same multiplicity.**

1.8 Remark — Three Cousins, Three Different Answers

This problem completes a trio that runs through Chapter 3, and it is worth seeing them side by side because the answers are all different:

statement	holds exactly for	reference
$n \mid m^2 \implies n \mid m$	n squarefree	3.3 P2, this problem
$n \mid bc \implies n \mid b \text{ or } n \mid c$	n prime (or 1)	3.4 P2
$n \mid m \implies n \mid m^2$	all n	3.3 P2, the easy direction

The middle row is the strictest because b and c are *independent* — any composite can be split as $n = uv$ and torn apart. In the top row the two factors of m^2 are forced equal, so only a repeated prime can cause trouble, and squarefree suffices. The bottom row is free: squaring can only increase exponents.

The modulus $n = 6$ separates the top two rows: it is squarefree, so $6 \mid m^2 \implies 6 \mid m$ holds, yet $6 \mid 2 \cdot 3$ while 6 divides neither factor. Being asked “does divisibility pass to the factors?” is a genuinely different question depending on whether the factors are equal.